

Ticket Master :

Functional requirements:

User should be able book tickets based on his preference,

User should be able to search for some particular events ,

User should be able to make payment

Non-functional requirements:

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here we need to prioritise Strong Consistency in order to avoid Double booking/Double Payment & for search scenario we need to prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:

user ,

ticket,

booking,

payment ,

history

API :

POST:user/booking

$$\{$$

userID,

```
{FileMetadata}
```

}

GET:/search/{eventID}

HLD :

